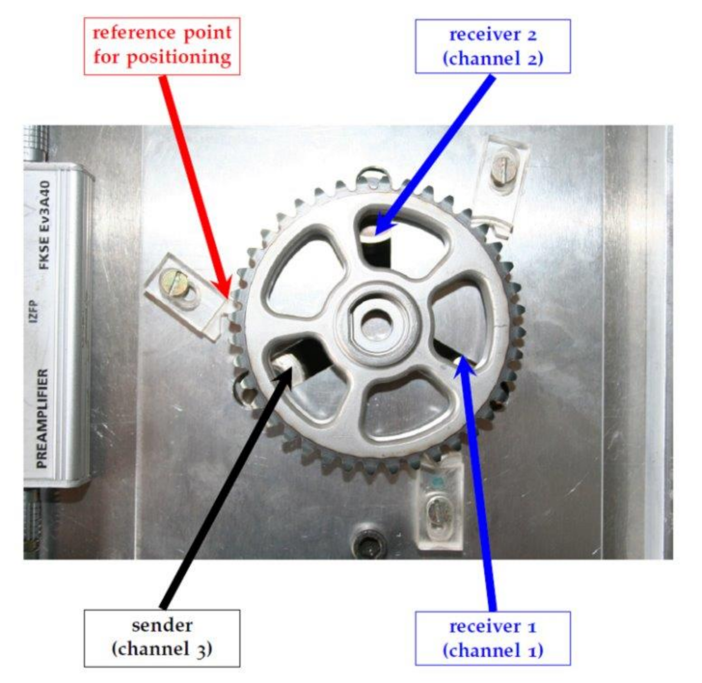
EXPLORATORY ANALYSIS OF NDT DATA FROM GEARS



Date: 19th June 2025

Subject: Artificial Intelligence in Material Diagnostics

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OUTLINE

1. Introduction and task
2. Data set and loading of data
3. Feature analysis of the signals
4. Principal Component Analysis (PCA) for dimensionality reduction
5. Visualization and evaluation

1. INTRODUCTION AND TASK

Explorative analysis is crusial for finding pattern and trends in data and also for detecting outliers.

Non-destructive testing is important to keep gears (or other materials) intact.

Task:

- Load the given signals into a data frame
- Apply principle component analysis (PCA) to reduce the dimensionality of the features
- Visualize the data points in 2-dimensional diagrams
- Evaluate the results

2. DATA

- Data set: 5 gears (Z01–Z05)
- 110 position files (Z01: 50, Z04: 30, others: 10 each)
- Each position file: 2 channels with 2 variants each
- In total: 440 data files

2. DATA

- We use os.path.join for reading the data
- We get all the data with .wav at the end

```
# - Durchsucht Unterordner Z01 bis Z05 jeweils in deren Wav-Unterordner
pattern = os.path.join('*Datei', 'sig', 'Z*', 'Z*', 'Z*', 'Wav', '*.wav')

# Alle Dateien finden, die dem Muster entsprechen
files = glob.glob(pattern)
print(len(files), 'files', '\n', files[264])  # Print for Error-Checks
```

- In 'entry' we put the keys, according to the task

- After that we add it to our data list

```
    # Extrahiert Dateinamen
    basename = os.path.basename(fn)
    parts = basename.split('_')

    # Erstellt Datensatz mit Metadaten und Signal (Keys der Aufgabenstellung entsprechend)
    entry = {
        'fn': basename,
        'sig': signal,
        'spec': parts[0],
        'mID': parts[4],
        'time': parts[5],
        'rID': parts[6],
        'sID': parts[8][:-4],
        'stft': None
    }

    data.append(entry)  # Fügt Datensatz zur Liste hinzu

# Erstellt DataFrame aus der Liste
df = pd.DataFrame(data)

# Zeigt die ersten 5 Einträge zur Qualitätskontrolle (Error-Checking)
print("DataFrame-Übersicht:")
```

3. FEATURE ANALYSIS OF THE SIGNALS

We extract the following features:

- Magnitude Mean: Magnitude by averaged time
- Spectral mean: average frequency, weightened by magnitude
- Spectral variance: distance of each frequency bin from the mean
- Upper entropy: describes how irregular the upper spectrum part is
- Flatness: used for finding out how much noise there is
- Rolloff frequency: 85% of energy is below this frequency
- Upper kurtosis: reflects spikyness or asymmetry in the upper hand

3. FEATURE ANALYSIS OF THE SIGNALS

More signal features:

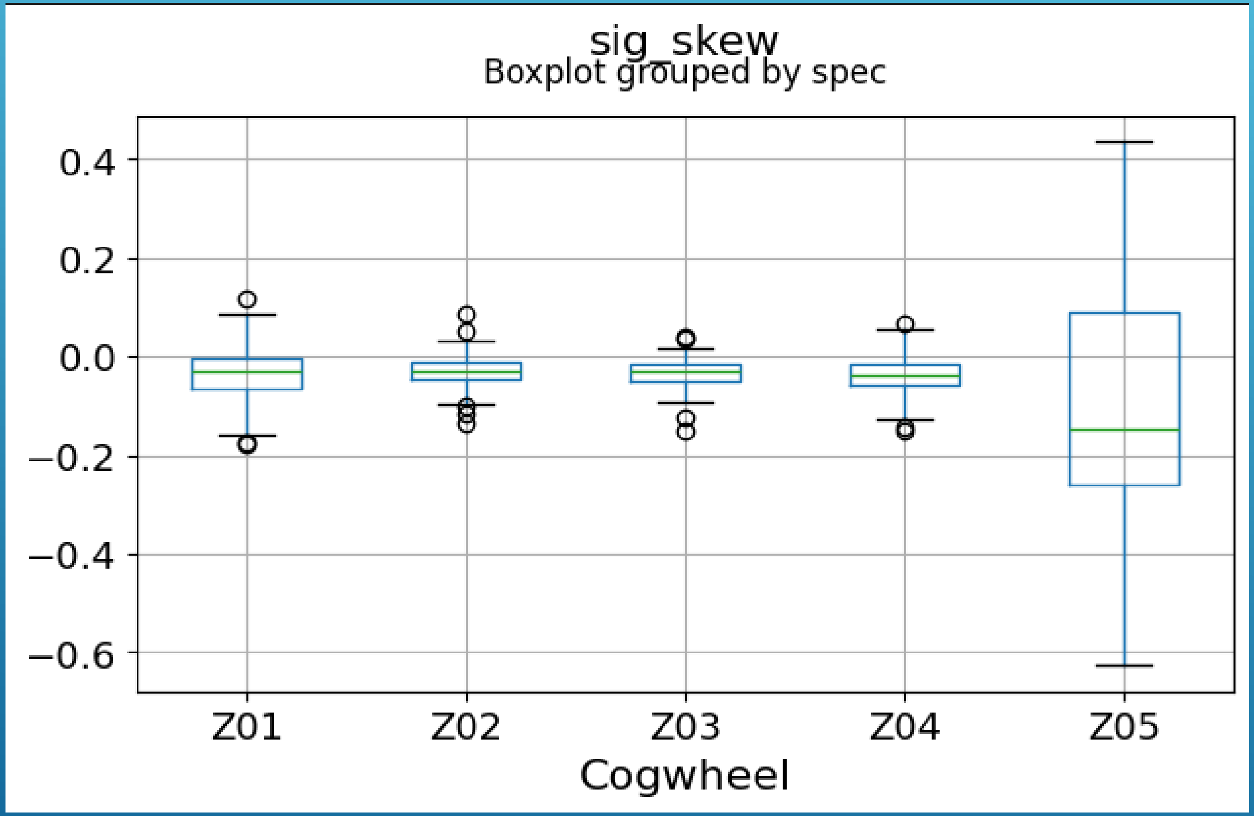
```
def analyze_signal(sig):
    signal_mean = np.mean(sig)
    signal_variance = np.var(sig)
    signal_skewness = skew(sig)
    signal_kurtosis = kurtosis(sig)  # fourth moment
    signal_rms = np.sqrt(np.mean(sig**2))
    signal_peak = np.max(np.abs(sig))  # peak amplitude
    return signal_mean, signal_variance, signal_skewness, signal_rms, signal_peak, signal_kurtosis
```

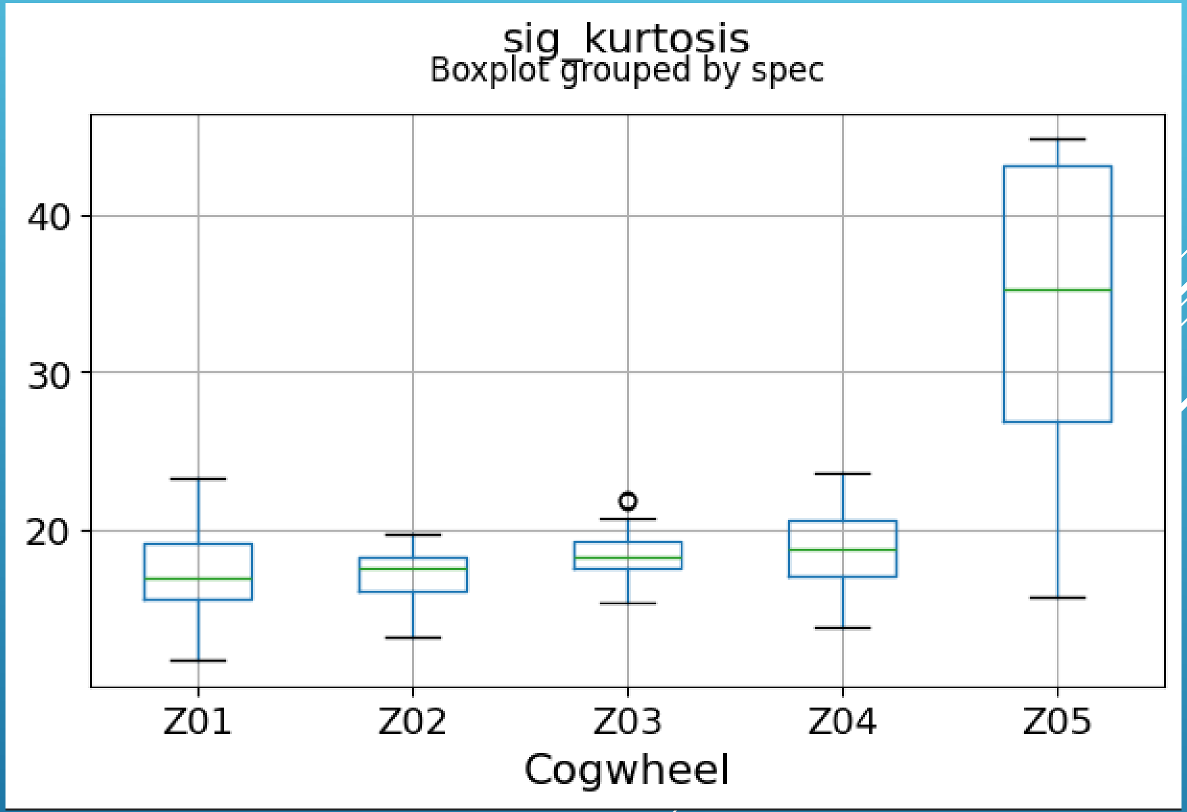
The feature list:

```
featureslist = ['sig_mean', 'sig_var', 'sig_skew', 'sig_rms', 'sig_peak',
'sig_kurtosis' 'spec_mean', 'spec_var', 'spec_entropy_up', 'spec_flatness',
'spec_rolloff_freq', 'spec_kurtosis']  # added new features
```

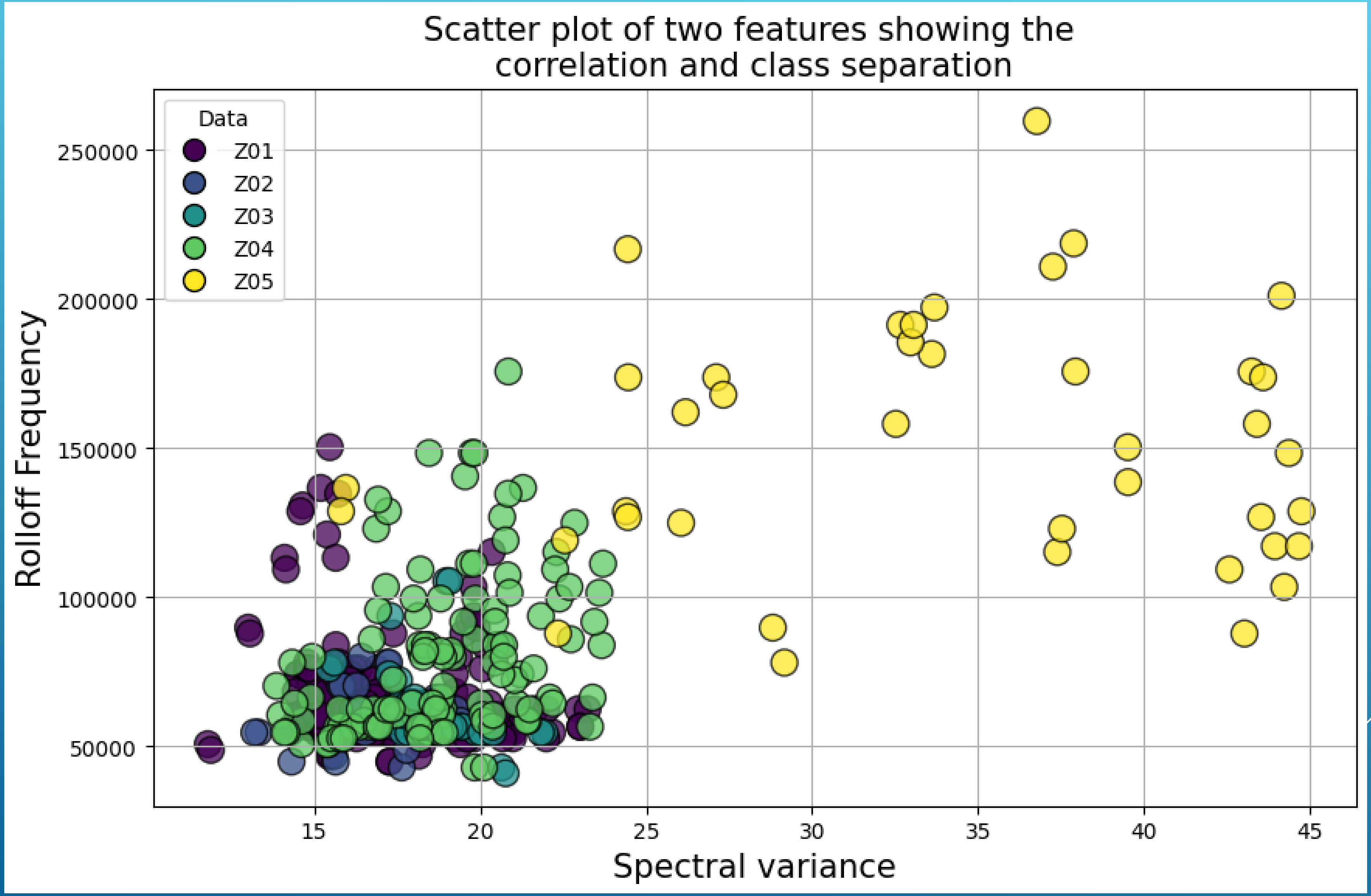
3. FEATURE ANALYSIS OF THE SIGNALS

Boxplots for skew and kurtosis:



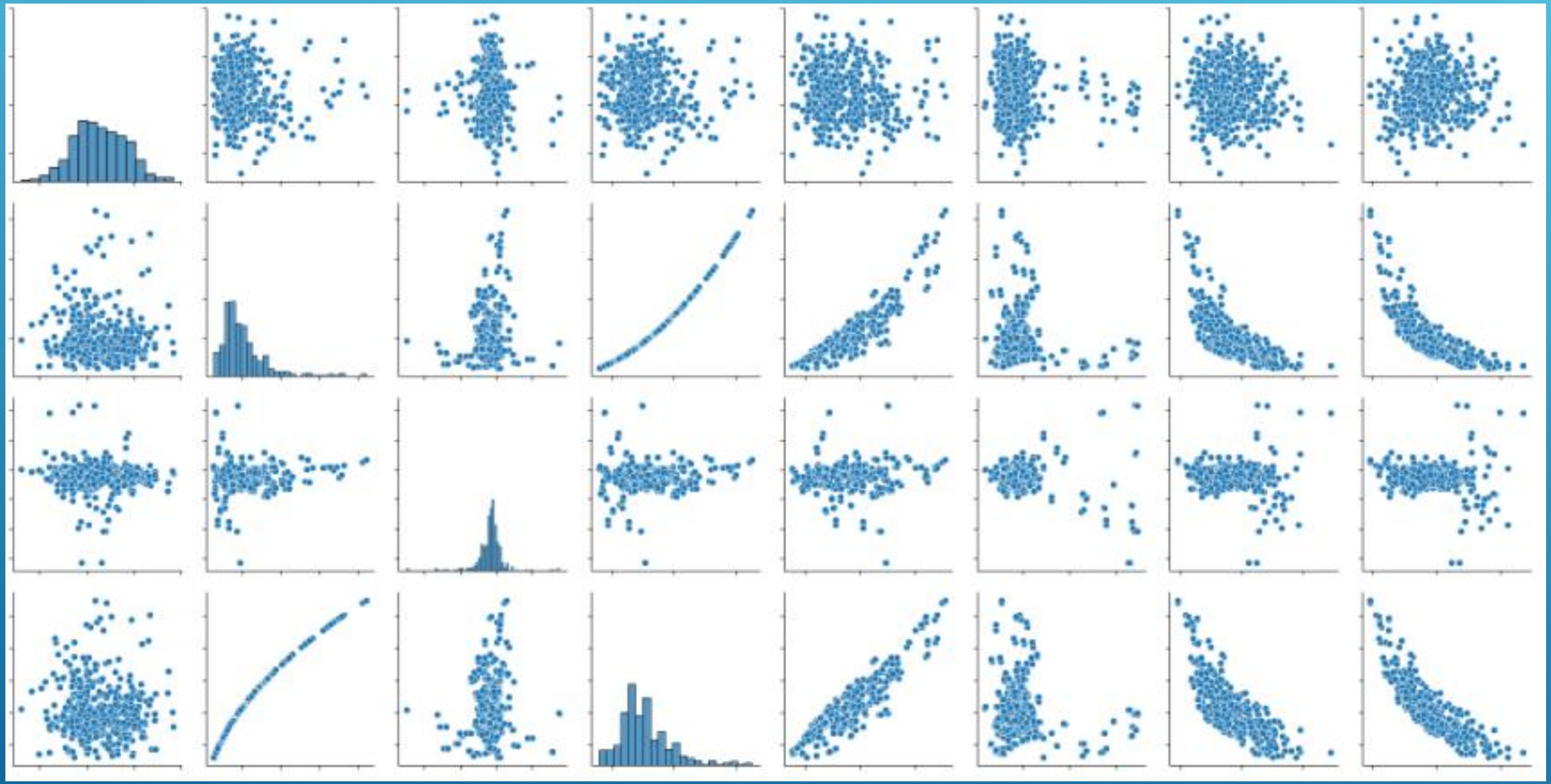


3. FEATURE ANALYSIS OF THE SIGNALS



3. FEATURE ANALYSIS OF THE SIGNALS

Feature Overview using Seaborn:



4. PRINCIPAL COMPONENT ANALYSIS

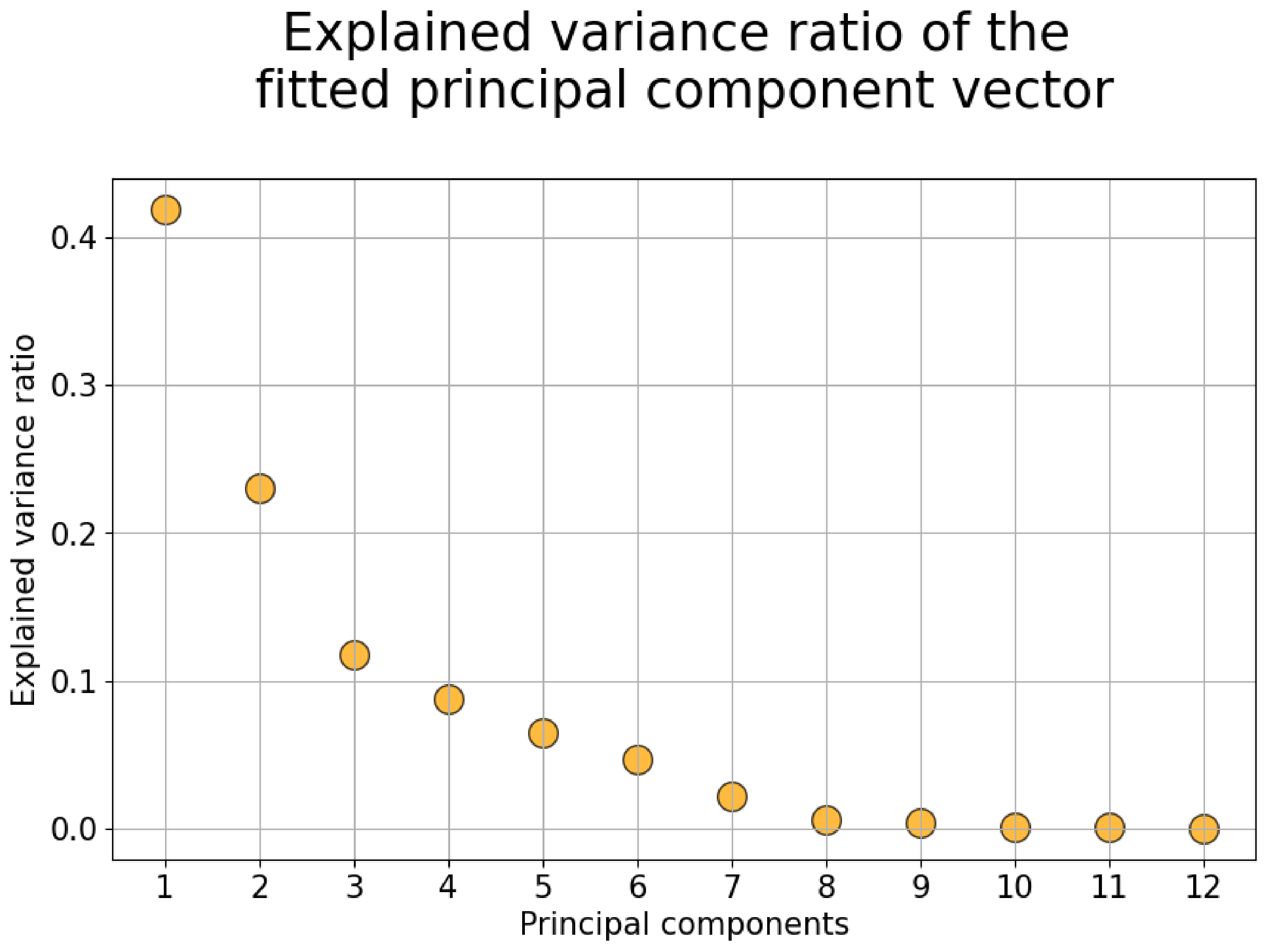
```
pca = PCA(n_components=None)
dfx_pca = pca.fit(dfx)
```

Goal of PCA: Reduce the number of variables by using new uncorrelated components that explain as much variance in the original data as possible.

- PCA calculates linear combinations of the original variables, where the first principal component captures the greatest variance

- The *explained variance ratio* shows how much of the total variance is explained by each principal component

4. PRINCIPLE COMPONENT ANALYSIS



4. DIMENSIONALITY REDUCTION

Goal of dimensionality reduction: Reduce the number of variables in a data set without losing important information.

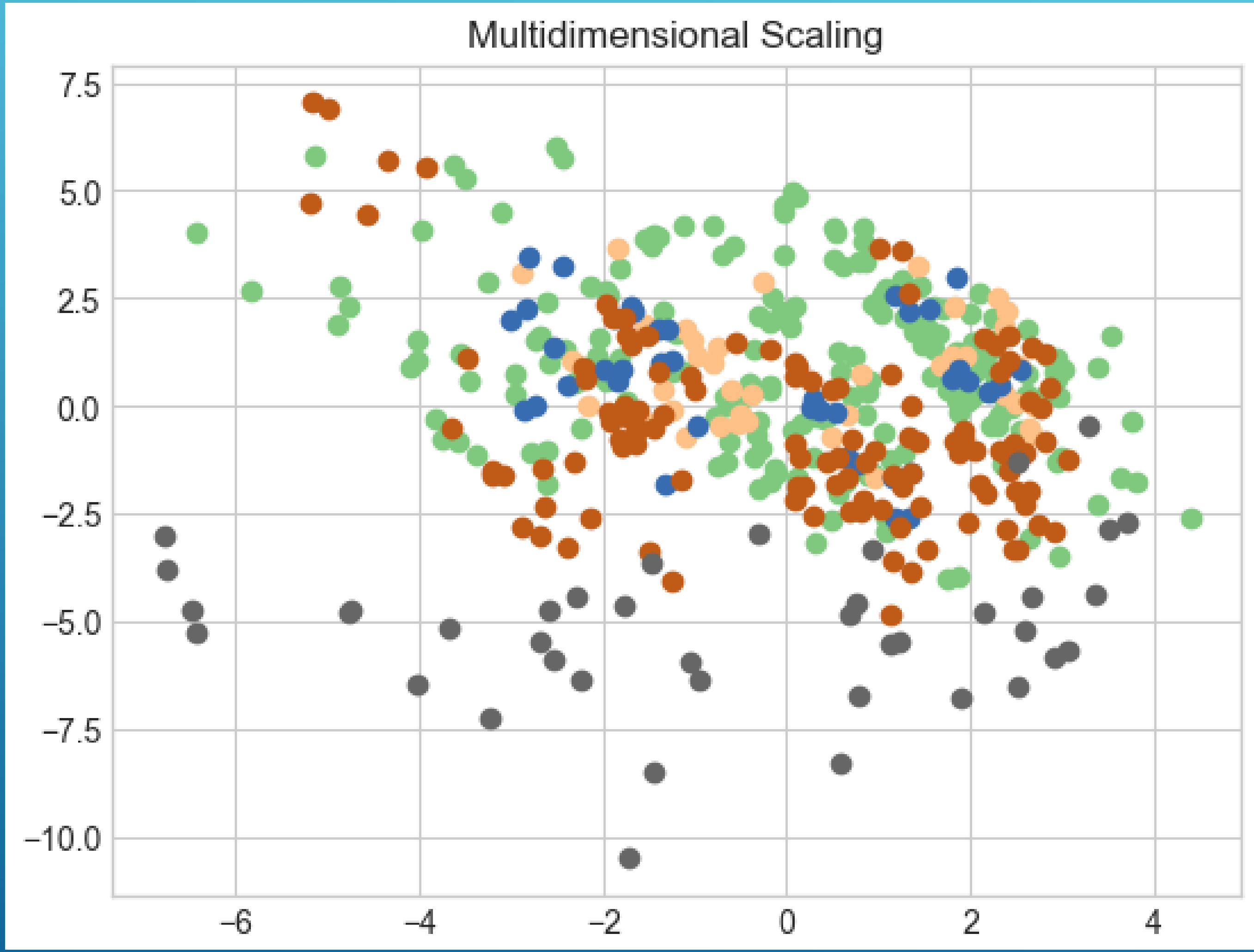
- We plot 2D and 3D scatter plots

```
with plt.style.context(plt_style):
    fig = plt.figure()
    if dimension == 2:
        ax = fig.add_subplot(111)
        for lab, col in zip(label_types, colors):
            ax.scatter(X_fit[labels==lab, 0],
                       X_fit[labels==lab, 1],
                       color=col)
    elif dimension == 3:
        ax = fig.add_subplot(111, projection='3d')
        for lab, col in zip(label_types, colors):
            ax.scatter(X_fit[labels==lab, 0],
                       X_fit[labels==lab, 1],
                       X_fit[labels==lab, 2],
                       color=col)
```

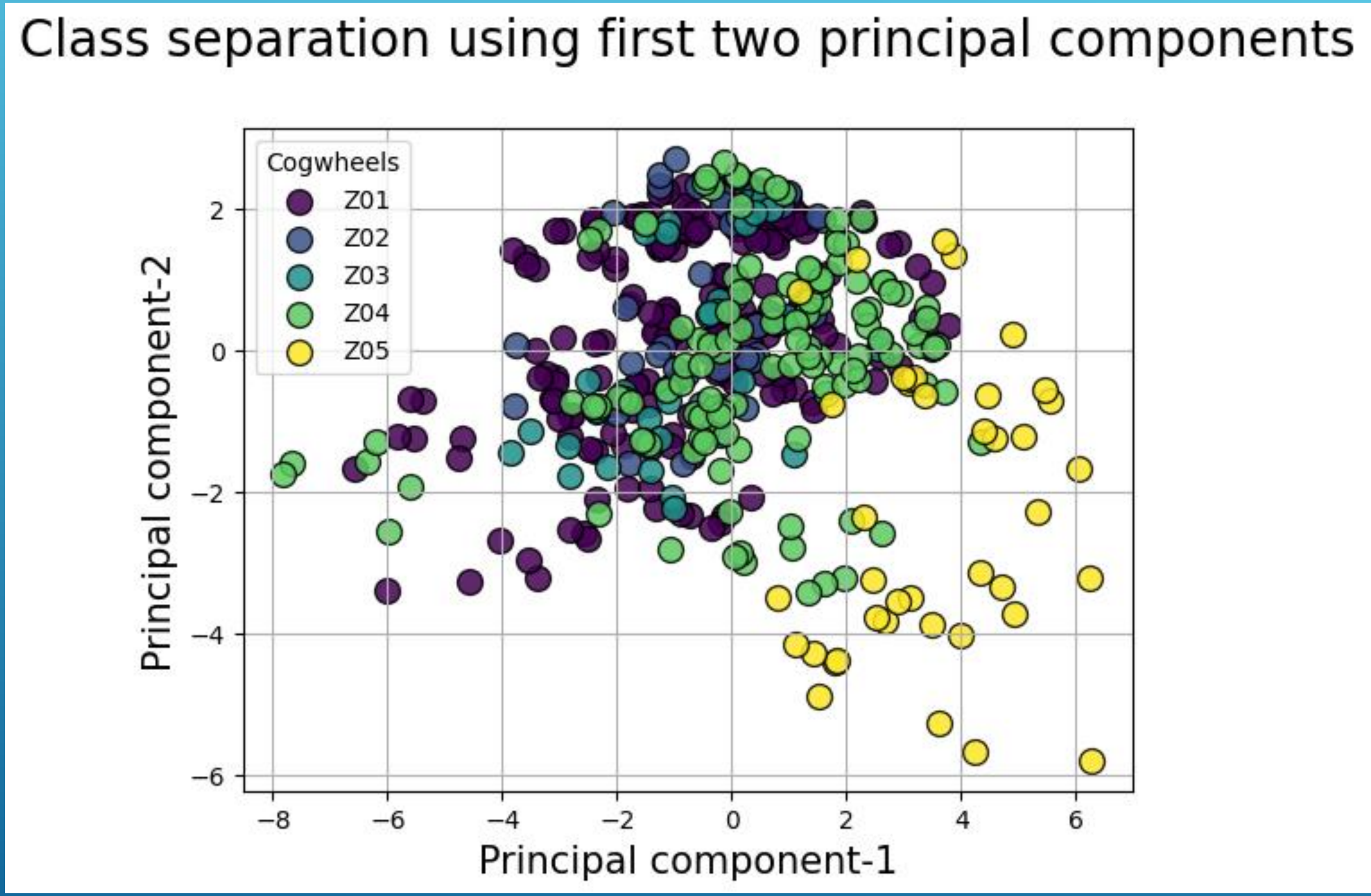
- Reduce dimensionality:

```
do_plot(PCA(n_components=2).fit_transform(dfx), 'PCA', df['spec'].values)
```

5. VISUALIZATION OF DATA POINTS



5. VISUALIZATION OF DATA POINTS



5. EVALUATION

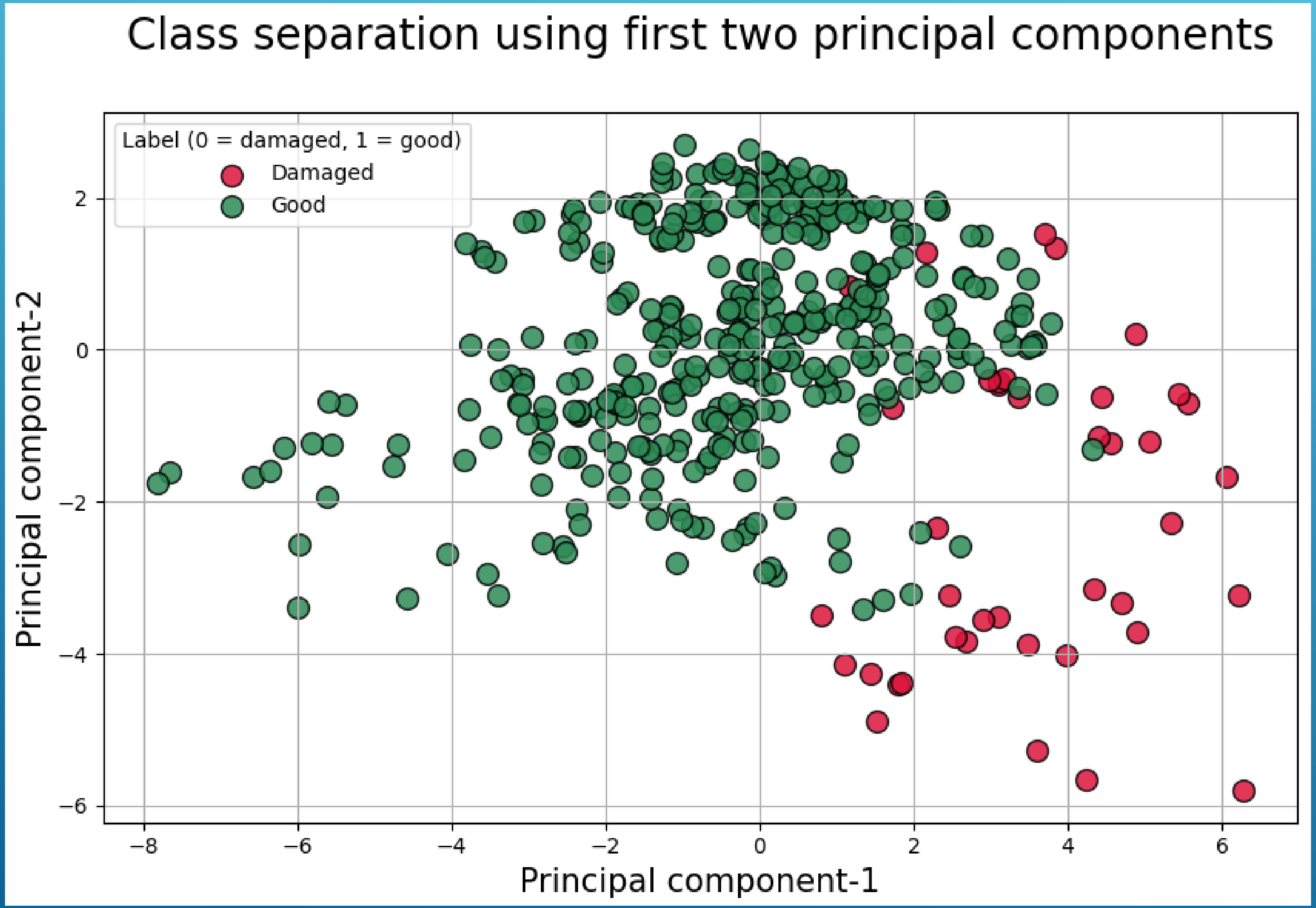
- We see that the data points appear around (0,0) in the diagram

- Z01 – Z04 appear similar to each other, because most of their data points are close to (0,0)

- Z05 is offset compared to Z01 - Z04

5. EVALUATION

As a result we create a class separation diagram:



5. EVALUATION

- We see that data points of Z01 – Z04 appear good
- Whereas data points of Z05 appear damaged

We conclude:

- Z05 is a damaged gear, while all others are in good condition
- But no gear is perfect! If one were, all of its data points would be at (0,0)

THANK YOU FOR YOUR ATTENTION